

Interpretable Medical Image Diagnosis Methodology using Convolutional Neural Networks and Bayesian Networks

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Abstract—A novel general methodology for fully interpretable AI concerning medical images is analyzed. The various consecutive steps of the method are presented in an instructive way. The algorithm is quite general, covering most of the cases of medical images. Interpretable diagnosis based on medical images is particularly useful for demanding applications like medical ones. Results show an efficient performance.

Keywords—AI medical diagnosis, Bayesian Networks, Interpretable AI, XAI.

I. INTRODUCTION

Explainable Artificial Intelligence (XAI) refers to the suite of techniques and methods that enable human users to understand and trust the results and outputs of machine learning algorithms. XAI aims to make AI decisions transparent, interpretable, and comprehensible, which is critical for fostering trust, ensuring accountability, and meeting regulatory requirements.

There are many applications of XAI. In Healthcare XAI is vital, ensuring that AI systems used in diagnostics and treatment recommendations are transparent and interpretable to healthcare professionals. Other fields where XAI is needed include, (i) Finance, providing clear explanations for credit scoring, fraud detection, and algorithmic trading decisions to meet regulatory requirements. (ii) Law, ensuring that AI systems used in legal decisions, such as sentencing and parole recommendations, are transparent and justifiable. (iii) Automated Systems: enhancing trust in autonomous systems, such as self-driving cars, by providing understandable explanations for their actions and decisions.

Advantages of XAI include: **Transparency**: Making the decision-making process of AI systems visible and understandable to humans. **Interpretability**: Ensuring that the model's predictions and internal workings can be easily interpreted by users. **Trustworthiness**: Building user

confidence in AI systems by providing clear, rational explanations. **Accountability**: Ensuring that AI systems can be held accountable for their decisions, particularly in critical applications.

Since these goals are quite ambitious XAI methods differ significantly and are under rapid progress. Some approaches to XAI are : Rule-Based Systems: These systems use explicit rules to make decisions, offering straightforward interpretability. LIME (Local Interpretable Model-agnostic Explanations), [1], LIME approximates the original model locally with a simpler, interpretable model (such as a linear model) to explain individual predictions. SHAP (SHapley Additive exPlanations), [2], SHAP values provide a consistent measure of feature importance based on cooperative game theory, offering both local and global explanations. Partial Dependence Plots (PDPs), [3], PDPs illustrate the relationship between a single feature and the predicted outcome, keeping other features constant. Individual Conditional Expectation (ICE) Plots, [4], ICE plots show the dependence of the prediction on a feature for individual instances. Feature Importance Plots, [5], Visualize the importance of each feature in the model's predictions. Attention Maps, [6], Highlight which parts of the input data the model is focusing on when making a decision, commonly used in image and text analysis. Saliency Maps: Used primarily in neural networks, these maps show which parts of an input image are most important for prediction. Counterfactual Explanations: Show how changing certain inputs can alter the output, helping users understand the decision boundary of the model.

-AI not only enhances the efficiency of data analysis but also opens new avenues for discovery by identifying patterns and correlations that might not be apparent to human observers. Explainable AI (XAI) is a rapidly evolving field in artificial intelligence that aims to make AI systems more transparent, understandable, and trustworthy. Despite its challenges, the field of XAI holds immense potential for making AI stronger and accessible, ultimately leading to more robust, fair, and accountable AI systems.

Elaborating on the methods for Explainable AI (XAI)

involves diving deeper into the various techniques and approaches that make AI models more understandable and interpretable. These methods can be broadly categorized into inherently interpretable models, post-hoc interpretation techniques, visualization tools, and interactive explanations. Interpretable and explainable AI using Bayesian networks involves leveraging the probabilistic and graphical nature of these networks to provide insights into AI decision-making processes. Bayesian Networks (BNs) are a type of graphical model that represent the probabilistic relationships among a set of variables.

Bayesian networks offer a robust framework for dealing with complex systems where uncertainty and probabilistic reasoning are key. Their ability to model causal relationships and integrate various data sources and expert knowledge makes them a powerful tool in various fields. However, the challenges related to scalability, data requirements, and network learning highlight the need for advanced computational methods and careful application. As computational power continues to grow and more sophisticated algorithms are developed, the potential applications and effectiveness of Bayesian networks expand, making them even more valuable for decision-making in uncertain environments [7-19]. Bayesian networks allow for the easy integration of expert knowledge and data, and the network can be updated as new information becomes available. The directed edges in Bayesian networks can often be interpreted as causal relationships, which is useful in many applications [7-19].

Bayesian Networks contribute to interpretability and explainability in AI. Bayesian networks are visual graphs where nodes represent variables, and edges represent probabilistic dependencies between these variables. This structure makes it easier to see and understand how different variables influence each other. The graph-ical nature of Bayesian networks can be more intuitive than complex, black-box models like deep neural networks, especially for users with less technical background. Bayesian networks use conditional probabilities to describe the relationships between variables. These probabilities can be used to explain the reasoning behind predictions or decisions made by the network. One of the key strengths of Bayesian networks is their ability to model causal relationships. Understanding these causal relationships can be critical in making AI systems transparent and their decisions explainable. Furthermore, Bayesian networks allow for what-if scenarios, where one can manipulate the variables to see how changes affect other parts of the network, thereby providing insights into causal effects. Bayesian networks can update their beliefs (probability distributions) in light of new evidence, which is a transparent process that can be tracked and understood. They also allow for the integration of prior knowledge through prior probability distributions. This feature can make the model's reasoning process more transparent and relatable, especially when the prior knowledge is domain-specific and interpretable.

The aim of this study is to present a novel XAI method that integrates BNs and CNNs to build an expert system that not only makes a medical diagnosis but also explains the decision providing the crucial and critical factors that drive the classification. Furthermore, we demonstrate the applicability and feasibility of this method with a specific application.

II. METHODOLOGY

A. Integration of CNNs and BNs

For image recognition there are already several methods [20-26] that capture the critical parts of an image used for a classification with the help of CNNs. There are six specific post-hoc methodologies - Grad-CAM, LIME, XRAI, Shapley Additive exPlanations (SHAP), Smoothgrad, and Vanilla Gradients - along with an inherently interpretable model, TT. These post-hoc methods can be used on CNN models that showed the best performance in each dataset, while TT should be trained from the ground up on each dataset. All these methods offer interpretative insights through saliency maps, and TT additionally provides bounding boxes to accurately highlight the areas of interest. The selection of these methods was based on their ability to make CNN models interpretable, bypassing the need for training on specially annotated datasets for interpretability, such as those labeled with specific image-concept annotations. Our proposed method is different from the above attempts and is explained below.

The aim is to understand the decision-making process of deep learning model. This is critical for scientific validity, ensuring that models generalize well to new, unseen data and are robust against variations in data quality and observation conditions. It is necessary to combine machine learning techniques with traditional physical models to enhance understanding and ensure consistency with established scientific knowledge. For accomplishing these tasks we will use Convolutional Neural Networks, [20-30], and specially designed algorithm that is able to identify the critical parts of the image that play the crucial role of the CNN classification. The next step is to define several important statistical computational quantities that estimate various characteristics of the selected areas of the images, see for an extensive list of these quantities in the following subsections.

All these statistical and computational quantities will be the nodes of a Bayesian network that will be trained and finally will be able to predict the causal influences of the significant factors that determine the final decision/classification with similar accuracy of the CNNs that have been used initially, see [18,19]. Furthermore, the BN not only will encapsulate these factors from the critical parts of the images but also will include available factors from the medical record of the patients. The latter is very important and generates a final diagnosis based both on the images and the patient's medical history.

Thus, the developed system will be able to successfully handle image data, making them suitable for analyzing medical images in an interpretable way. For example, with the help of a BN, the system will be able to uncover critical features related to the shape, size, color, temperatures, shadings of tumors or

other abnormalities, which are causally connected with the diagnosis together with personal information like age, results of tests etc.

B. Identifying critical for the classification/diagnosis parts of the Images

The steps of the algorithm are the following

1. Obtain a dataset of images
2. Train a CNN for diagnosis/classification
3. Use the trained CNN for diagnosis of one image from a new set of images or from the previous initial dataset
4. Split this image of the dataset into N parts (in the same way for all images)
5. Repeat from $i=1$ to N the following
 - Use the trained CNN to make diagnosis for the image without part i ($i=1 \dots N$)
 - If diagnosis remain the same then omit part i and continue to $i+1$ loop
 - Else keep part i and continue to $i+1$ loop
- Define the reduced image
- Evaluate factors from this reduced image
- Repeat the above for all images

If the XAI interpretation is considered very important for medical reasons, for example when a new imaging technique is under testing then a more precise and demanding algorithm can be followed :

1. Obtain a dataset of images
2. Train a CNN for diagnosis/classification
3. Estimate various metrics for the diagnosis
4. Use the trained CNN for diagnosis of one image from a new set of images or from the previous initial dataset
5. Split this image of the dataset into N parts (in the same way for all images)
6. Repeat from $j=1$ to N the following
 - Use the trained CNN to make diagnosis without part j ($j=1 \dots N$)
 - If diagnosis remain the same then omit part j and continue

○ Else keep part j and continue

7. Repeat previous loop content $N!-1$ times to cover all N-permutations of the enumerated parts
8. At most there will be $N!$ reduced images (not all of them necessarily different)
9. Evaluate factors from all the different reduced images that are alternatives to image j
10. Repeat the above for all images j

C. Factors

After finding the critical parts of the images, factors should be evaluated. These factors are statistical, computational quantities. Many statistical and computational quantities could be used. Depending on the type of medical images the most appropriate for the type of images under concern may be calculated. However, even if initially we will use an extensive list of factors this does not reduce the efficiency and accuracy of the overall algorithm (no overfitting problem) since BNs are driven by the most influential factors while at the same time can handle without problems less important factors. We can also visualize which factors have less contribution to the final diagnosis.

Examples of statistical or computational factors are Maximum, minimum, density of a specific color, variances, distances between minima maxima and mean values, other statistical distances, counts, occurrences of gradient orientation in localized portions of an image, Scale-Invariant Feature Transform (SIFT) Features, Speeded Up Robust Features (SURF), Zernike Moments, spatial localization, orientation, and frequency details of an image, color distribution characteristics, Edge Detection Metrics (e.g., Canny Edge Detector), Gabor filters, Measures for identifying significant edges in images, crucial for object detection and segmentation, Pixel Intensity and Contrast ratios, other statistical measures like entropy, homogeneity, perimeters of critical areas, eccentricities, circularity, GLCM, dissimilarity. These quantities will be the informational nodes of the Bayesian Network with target variable the classification decision.

Furthermore, BNs can include in the previous set of factors more informational nodes concerning medical factors from the patient's historical record. This is crucial for medical purposes. Examples of such factors are : Date of birth, Gender, Past illnesses and chronic conditions (e.g., diabetes, hypertension), Surgical history (dates and types of surgeries), Family medical history (health conditions prevalent in the family), Current and past medications (names, dosages, frequencies), Detailed descriptions of symptoms, diagnoses, and treatment plans, Blood tests, urine tests, and other lab results, Pathology reports (biopsy results, cytology), Referrals to other healthcare providers, Smoking, alcohol consumption, and drug use, Exercise habits and diet, Mental health status and any psychological evaluations.

D. BN

A Bayesian network is a directed acyclic graph (DAG), where each node represents a variable (which could be discrete or continuous), and edges represent conditional dependencies. The absence of an edge implies conditional independence between the variables. Each node in the network is associated with a Conditional Probability Table CPT that quantifies the effect of the parent nodes on the node. For a node with no parents, the CPT reduces to a prior probability distribution. A key strength of Bayesian networks is their ability to represent the joint probability distribution of all the variables in the network using the product of conditional probabilities as specified by the network structure.

The structure of a Bayesian network can be either determined by expert knowledge or learned from data like in our case. The structure encodes the conditional dependencies between variables. Once the structure is defined, the next step is to learn the parameters of the CPTs. This can be done using maximum likelihood estimation or Bayesian methods, often based on available data. The process of computing the posterior distribution of certain variables given evidence about others. Inference can be exact (using algorithms like variable elimination or belief propagation) or approximate (like Monte Carlo methods).

In our case the nodes will be the previously mentioned factors and the CPTs will be automatically estimated from the relevant frequencies of the factors estimated from all the images (the critical parts of them). The steps regarding the BN are:

11. Design BN with the statistical, computational, medical factors as informational nodes
12. Train BN for diagnosis
13. Evaluate metrics for the diagnosis
14. If the metrics are similar to the CNN (similarity defined by inequalities) then continue else stop
15. Add more factors
16. Evaluate metrics for the diagnosis
17. If the metrics are similar to the CNN (similarity defined by inequalities) then continue else stop
18. Evaluate the strength of most influential nodes for the diagnosis
19. Create list of significant nodes and their strength regarding the target variable that concerns the diagnosis states and of their interconnections.

E. The whole Algorithm

- A. Preparation of the appropriate dataset
 - a. Data Collection: Gather a comprehensive set of images relevant to the specific field of study.
 - b. Data Preprocessing: Clean and preprocess the images, which could include tasks like resizing, normalization, and augmentation to improve model training.

- c. Labeling and Annotation: Annotate the images with labels or relevant information, essential for supervised learning tasks.

B. Designing of CNN model optimal for the particular dataset.

- a. Architecture Design: Develop a CNN architecture that is well-suited for the types of images and features relevant to the analysis.

C. Training CNN and evaluation of it with relevant AI metrics

- a. Train the CNN on the prepared dataset, using state-of-art optimization methods and adaptive learning rate algorithms.

D. Running the trained CNN

E. Spot the critical parts of the images with the previously analysed algorithm in subsection B

F. Calculating several statistical and computational quantities that characterize the critical parts of the image. This might include color distribution, texture patterns, edge details, and other relevant image features.

G. Add in the set of factors the medical history factors, see subsection C.

H. Generating a Bayesian Network with informational nodes the previously mentioned factors

I. Training the BN, Adjust the BN's parameters and structure to accurately represent the relationships between different image features and patient's data..

J. Running the BN

K. Discovering the influential informational nodes for classification

- a. Node Analysis: Examine which nodes/factors in the Bayesian Network are most influential in the classification or prediction process.
 - b. Evaluate strengths of interconnections of factors.
 - c. Insight Extraction: Gain insights into which specific image features or medical data are most critical in determining the model's outputs, enhancing understanding and guiding future model improvements or medical data collection.

III. PRELIMINARY RESULTS

The full results will be reported in a forthcoming paper. Here we report some preliminary results and some issues concerning the implementation of the proposed methodology.

- We have used one dataset for medical images about pneumonia taken from NIH Chest X-ray Dataset.
- The first less demanding algorithm of subsection B was used for identifying the critical parts of every image
- The accuracy of the BN diagnosis was very close to the CNN around 86%

- The most influential nodes were: grey color density, computational quantities for branching, tubular structures, fine, linear or reticular opacities, angles of lines, whole lung volume (area), cavities, diffusion of grey, differences of above between lungs, and statistical distances.

IV. CONCLUSION

XAI is a critical area in AI research for medical applications, focusing on making AI systems transparent, interpretable, and trustworthy. By employing a variety of techniques and approaches, XAI aims to build user confidence, ensure accountability, and meet regulatory requirements, thereby enabling the responsible and ethical use of AI in various domains. Medical diagnosis with the help of machine learning techniques [31,32] is a very important research area that demands interpretable diagnosis.

In the present work we have developed a new methodology for interpretable diagnosis based on medical images.

Some challenges in XAI that constitute motivations for future research are: 1) Complexity vs. Interpretability is a serious issue. Balancing the complexity of powerful AI models with the need for interpretability is challenging. Complex models often perform better but are harder to explain. In our case though the understanding of a diagnosis is crucial for a physician. 2) Scalability: Applying XAI techniques to large-scale AI systems can be computationally intensive and resource-demanding. 3) Ethical and Social Implications of XAI systems. Transparent explanations can sometimes reveal sensitive information, raising privacy concerns

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